SCA UWB BASIC

Competency program – sca uwB

November - 2025

AGENDA

01
Ultra WideBand (UWB) Technology)

02
Ranging Technics

03
UWB PHY

04
Ranging Session Setup

05
MAC Time Grid

06
MAC Time Grid Synchronization

07
Hopping and Resource Management

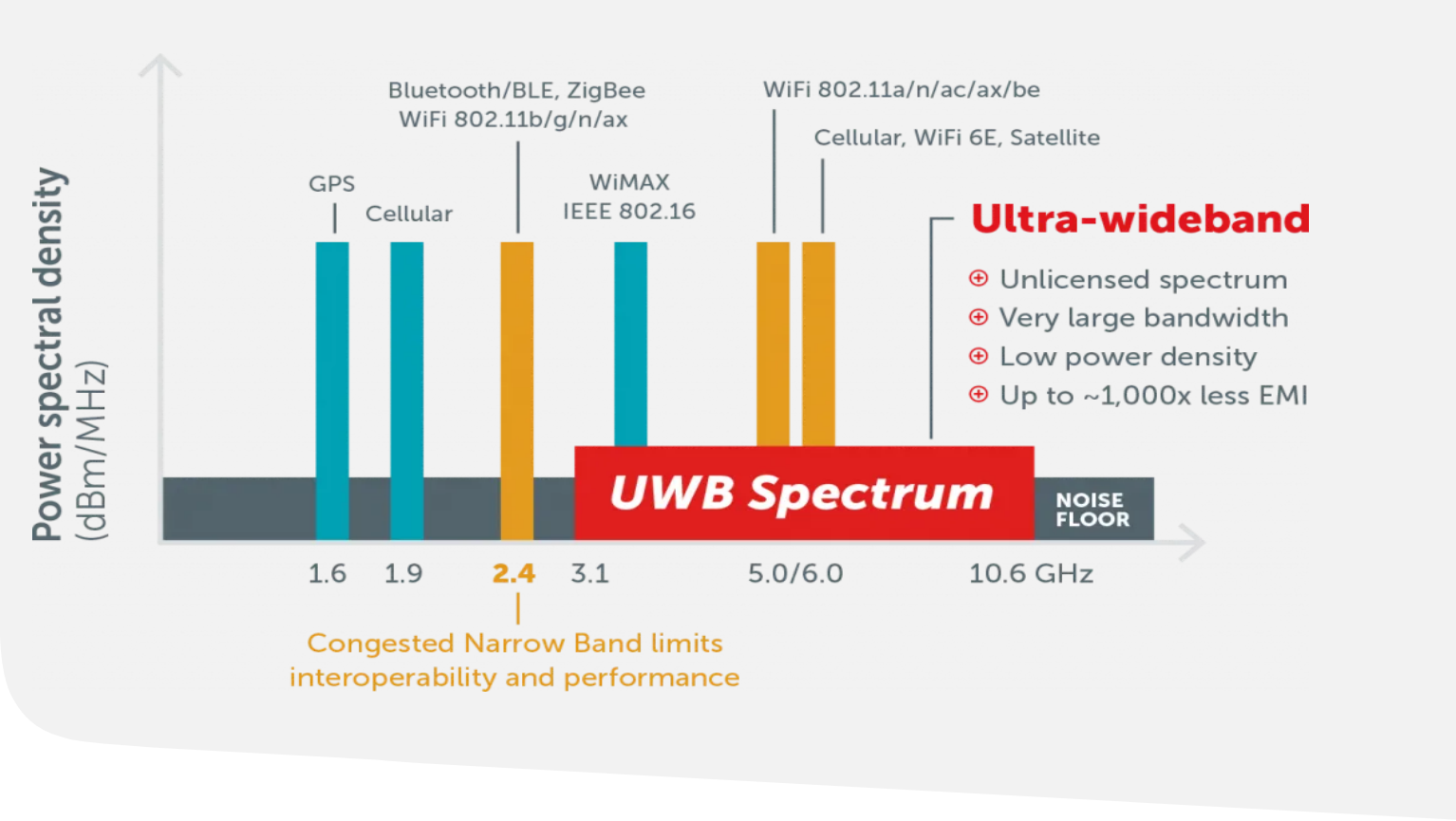
08
UWB Crypto Sequence

01
ULTRA WIDEBAND (UWB) TECHNOLOGY

ULTRA WIDEBAND (UWB) TECHNOLOGY

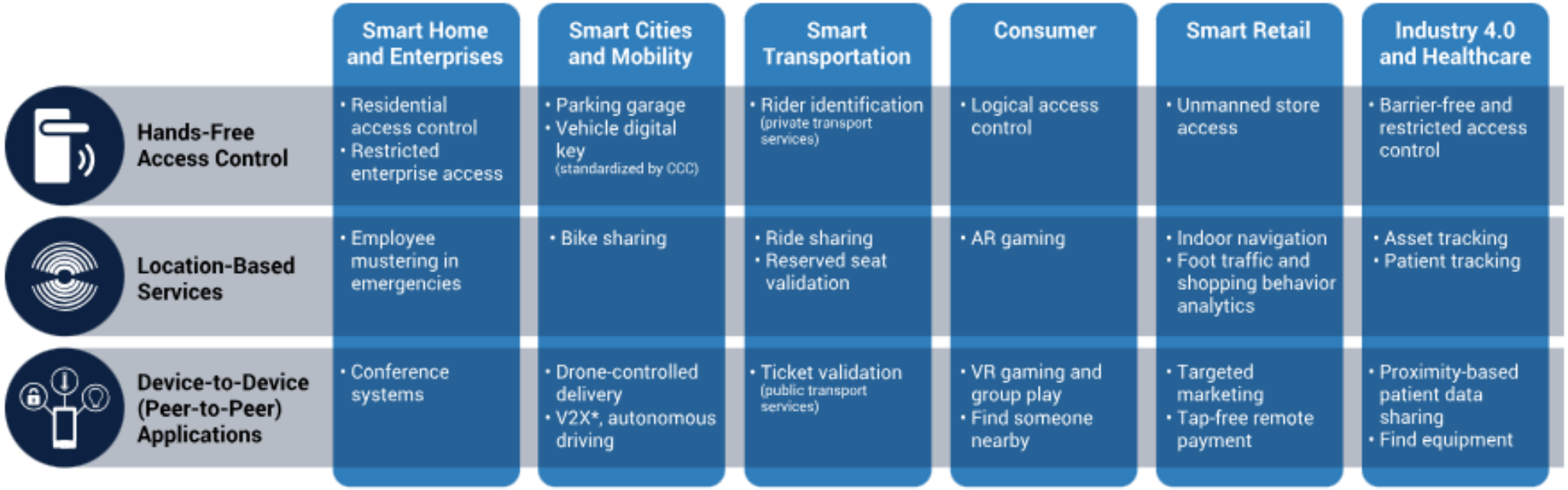
What's Ultra WideBand (UWB)?

- UWB stands for Ultra WideBand
- Short range wireless communication, from 3.1 GHz to 10.6 GHz with 16 RF channel(0-15)
- Low Power: very low energy level
- High Bandwidth (> 500 MHz), allowing for high data rates and precise locating tracking
- Main functions: indoor positioning, smart car access and smart home devices, …



ULTRA WIDEBAND (UWB) TECHNOLOGY

UWB Use-Case



	Smart Home and Enterprises	Smart Cities and Mobility	Smart Transportation	Consumer	Smart Retail	Industry 4.0 and Healthcare
Hands-Free Access Control	• Residential access control • Restricted enterprise access	• Parking garage • Vehicle digital key (standardized by CCC)	• Rider identification (private transport services)	• Logical access control	• Unmanned store access	• Barrier-free and restricted access control
Location-Based Services	• Employee mustering in emergencies	• Bike sharing	• Ride sharing • Reserved seat validation	• AR gaming	• Indoor navigation • Foot traffic and shopping behavior analytics	• Asset tracking • Patient tracking
Device-to-Device (Peer-to-Peer) Applications	• Conference systems	• Drone-controlled delivery • V2X*, autonomous driving	• Ticket validation (public transport services)	• VR gaming and group play • Find someone nearby	• Targeted marketing • Tap-free remote payment	• Proximity-based patient data sharing • Find equipment

*Connected Vehicle-to-Everything Communication

Smart Car Access - unlock a car with a smartphone as soon as you approach it for keyless entry and remote start.

ULTRA WIDEBAND (UWB) TECHNOLOGY

UWB & Bluetooth Comparison

Category	UWB	Bluetooth® Low Energy Direction Finding	Bluetooth® Low Energy RSSI	Bluetooth® Low Energy Channel Sounding
Ranging Method	ToF/AoA	AoA/AoD	RSSI	PBR/RTT
Ranging Measurement	Distance & Direction	Distance & Direction	Distance Only	Distance Only
Accuracy	< 10 cm	1-5 m	1-5 m	< 1 m (PBR) 1-5 m (RTT)
Range[1]	50 m typical	20 m typical	20 m typical	20 m typical (PBR) > 30 m typical (RTT)
Data Rate	Up to 27 Mbps	Up to 2 Mbps	Up to 2 Mbps	Up to 2 Mbps
Latency	1 ms typical	20 ms typical	20 ms typical	100 ms typical
Reliability	Strong immunity to multi-path and interference	Very sensitive to multi-path and interference	Very sensitive to multi-path and interference	Sensitive to multi-path and interference
Security	Very secure	Not secure, can be spoofed using relay attacks	Not secure, can be spoofed using relay attacks	Secure; anti-spoofing features included
Scalability	1,000's of tags to unlimited based on positioning scheme	1,000's of tags	1,000's of tags	Device to device, with potential to scale to 10's of tags

[1] Multiple factors impact range. For UWB, assumes line-of-sight, channel 5 (6.5 GHz) operation, 6.8 Mbps data rate, good antenna. For Bluetooth® Low Energy, assumes line-of-sight 1 Mbps data rate.

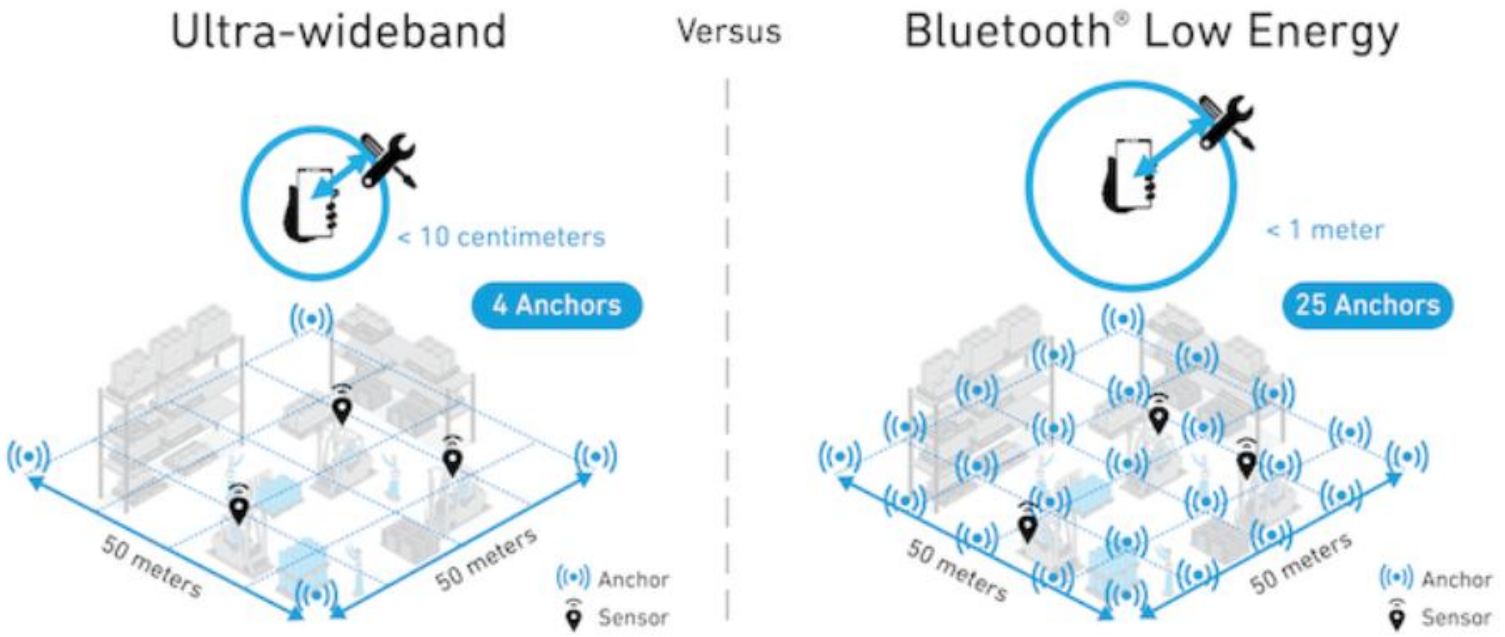
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The Future of Positioning: Exploring UWB and Bluetooth Channel Sounding - Qorvo

ULTRA WIDEBAND (UWB) TECHNOLOGY

UWB & Bluetooth Comparison



Sub-meter accuracy is achievable with both, but Bluetooth® Low Energy requires more anchors

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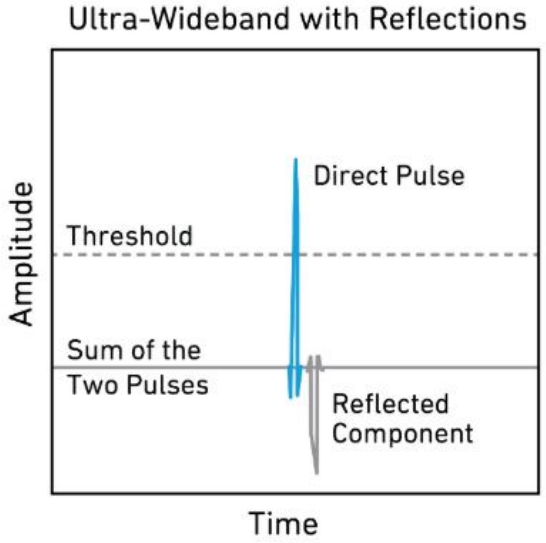
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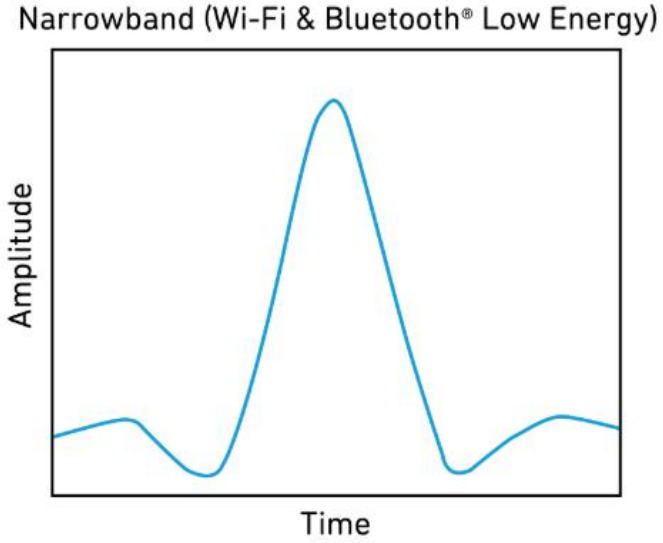
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ULTRA WIDEBAND (UWB) TECHNOLOGY

Security

- UWB signal (left) has a much faster rise and fall time than the narrowband signal shown on the right, enabling it to precisely measure the time of arrival of the signal, increasing its immunity to multipath effects and interference from other radios.
- By spreading its energy over a wide bandwidth and transmitting it at an extremely low power level of -41.3 dBm (MHz), UWB appears as wideband noise to other narrowband radio signals such as Bluetooth LE.





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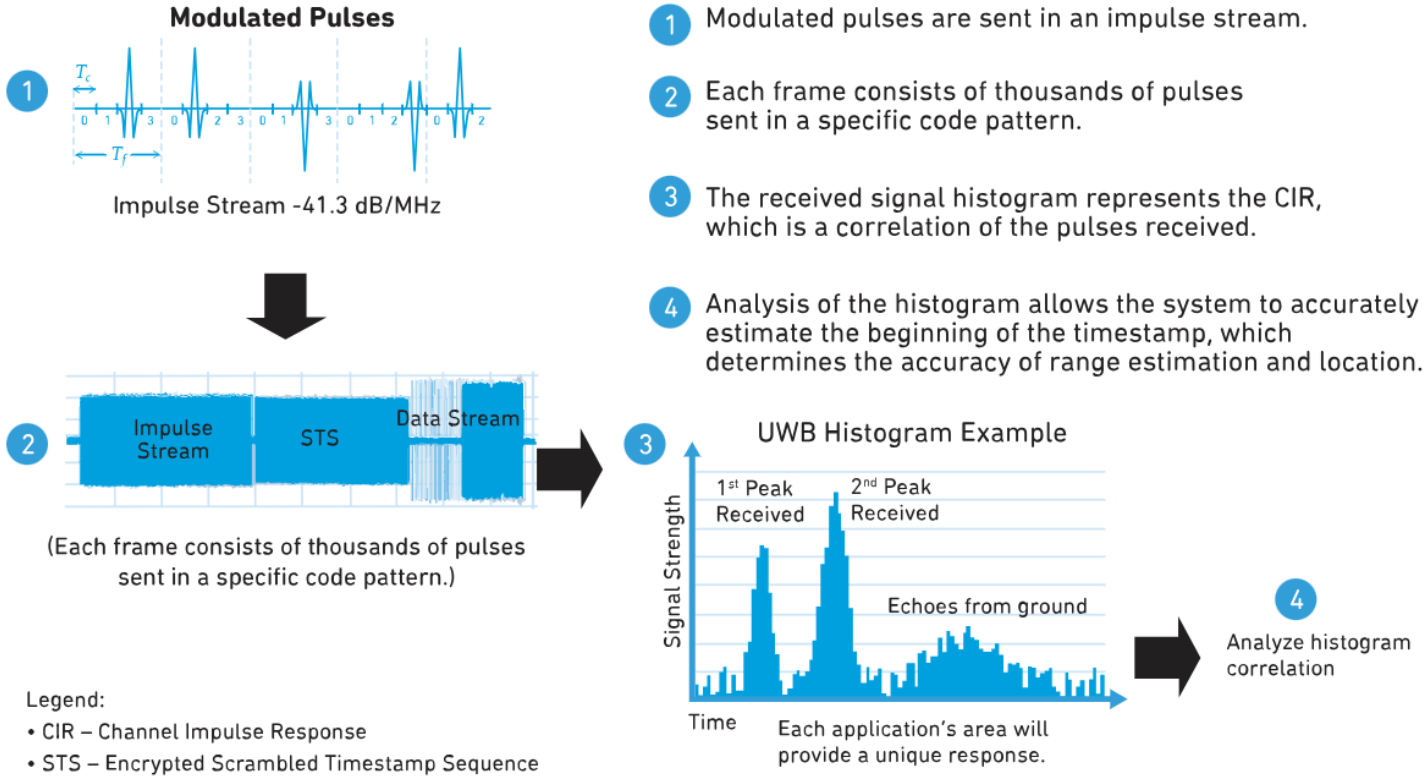
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ULTRA WIDEBAND (UWB) TECHNOLOGY

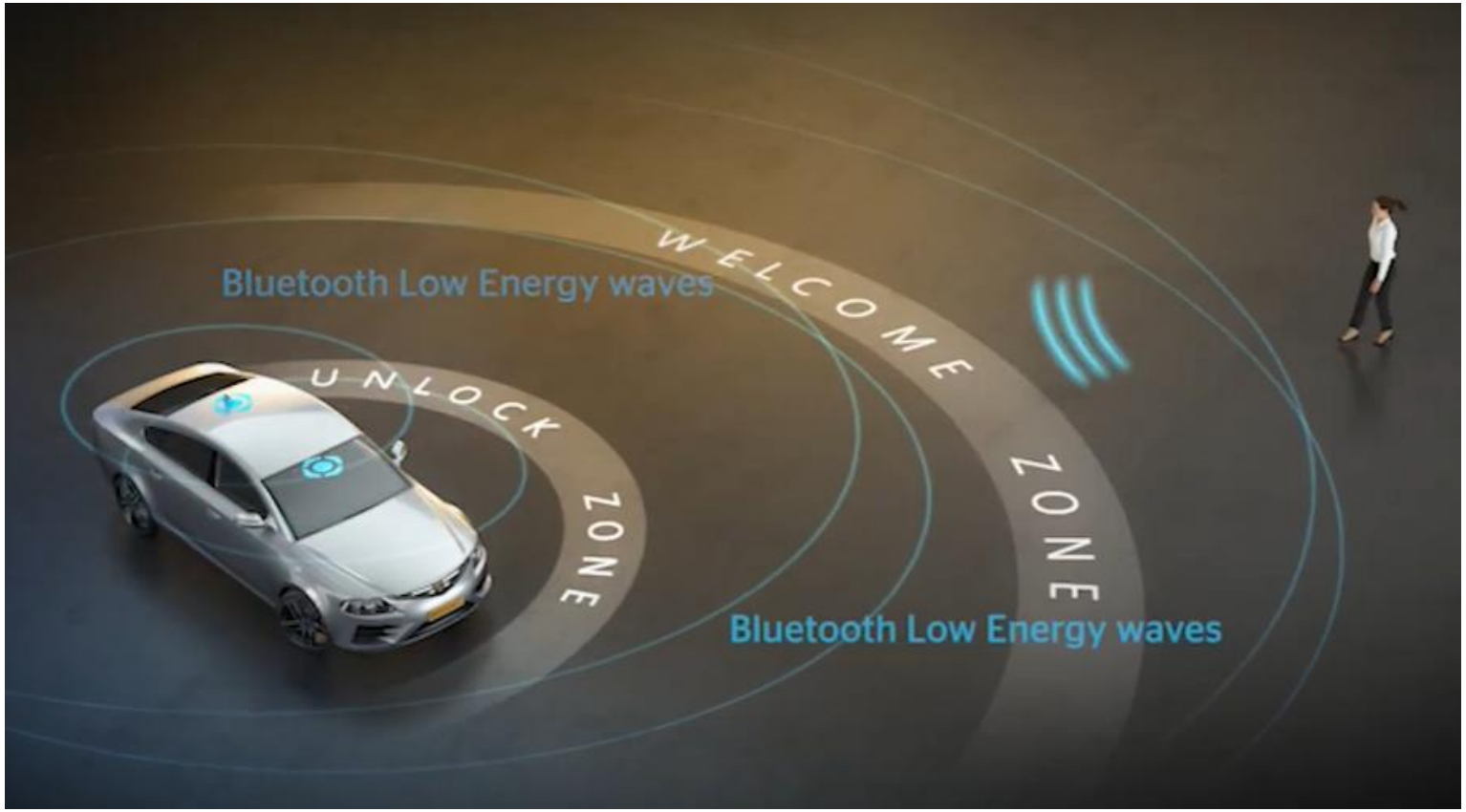
Security

- EEE 802.15.4z HRP UWB PHY adds a scrambled time stamp (STS) field into the packet. This STS field consists of a set of pseudo-random binary shift keying (BPSK) modulated pulses.
- The data cannot be accessed or predicted because cryptographic keys and random numbers are sent with the PHY packet



[Unlocking the Future: How UWB Is Reshaping Secure Access in the Connected World - Qorvo]

ULTRAWIDE BAND TECHNOLOGY



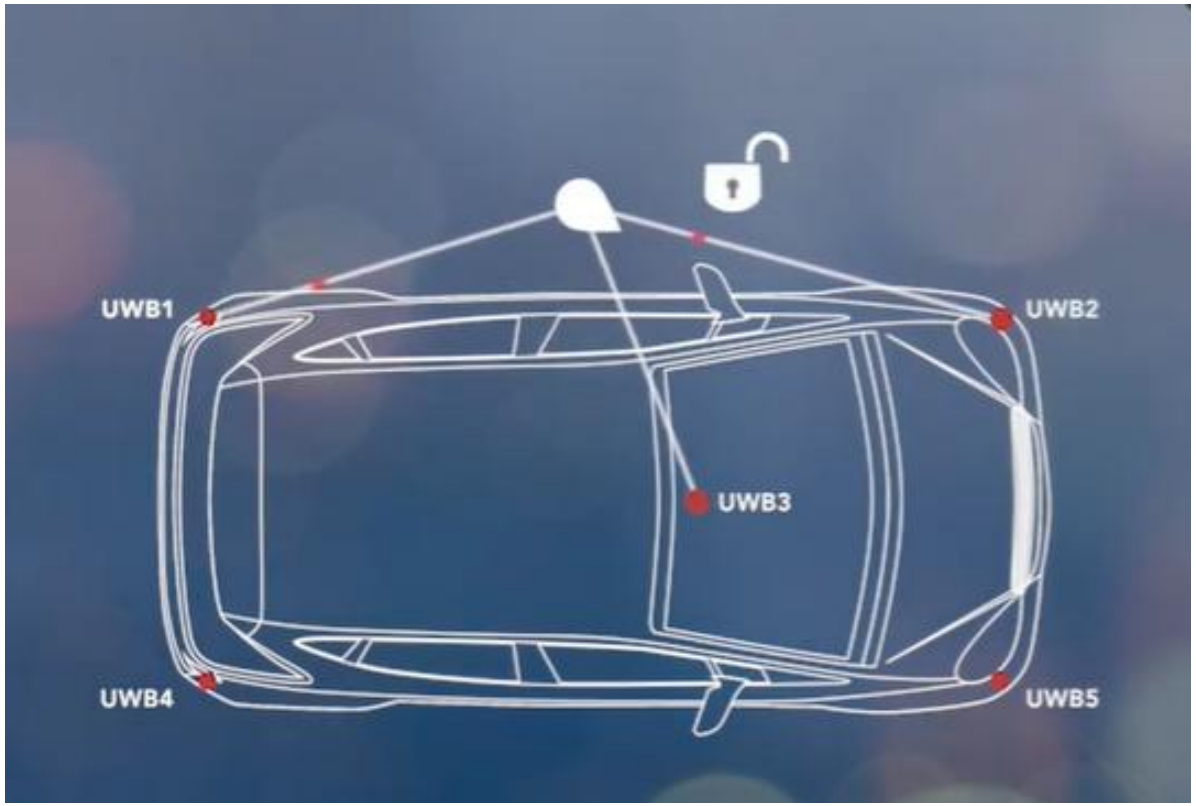
- BLE wakeup and established
- User authentication
- Credential exchange

ULTRAWIDE BAND TECHNOLOGY



- UWB synchronization
- Secure ranging

ULTRAWIDE BAND TECHNOLOGY



- Distance estimation
- Angle of arrival
- Door lock/unlock

02
RANGING TECHNICS

RANGING TECHNICS

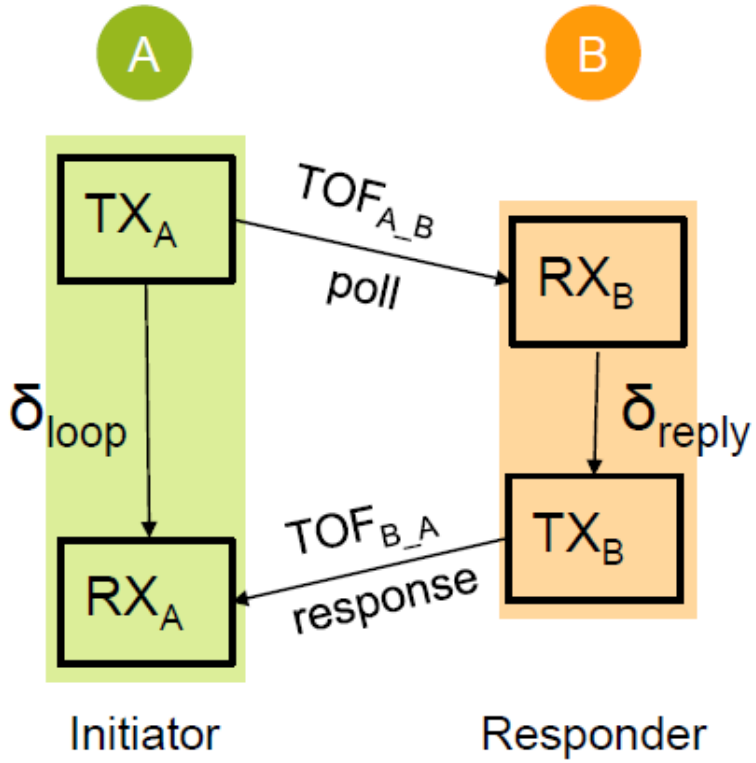
Ranging Roles Definitions

	Role	Function	Responsibility	Example in SCA
Initiator	The initiator is the device that starts the ranging process	It sends out a ranging request to the responder	Ensures accurate timing and synchronization to measure the distance to the responder	
Responder	The responder is the device that receives the ranging request from the initiator	It processes the request and sends back a response	Provides accurate timing information to help the initiator calculate the distance	
Anchor	Anchors are fixed reference points in a UWB network	They provide known positions to assist in location tracking	Maintain precise location information to help mobile devices determine their position	
Tag	Tags are mobile devices that need to be located	They communicate with anchors to determine their position.	Send and receive ranging signals to calculate their location relative to the anchors	

RANGING TECHNICS

Single Sided – Two Way Ranging (SS-TWR)

- **Time of Flight (ToF)** is measurement of the time it takes for the radio waves to travel a distance in the air
- **Initiator** (A) send a **poll** to **Responder** (B)
- Time needed by the **poll** to cross the distance D_{A_B} between A&B and reach B, is $\mathbf{ToF_{A_B}} = D_{A_B}/c$
- Delay of processing (between request receiving and response sending) $\boldsymbol{\delta_{reply}}$ is added in the PSDU response
- Delay needed by the **response** to reach A, is $\mathbf{ToF_{B_A}} = D_{B_A}/c = \mathbf{ToF_{A_B}}$
- Initiator knows delay between the poll sending and response receiving): $\boldsymbol{\delta_{loop}} = 2 \times ToF_{A_B} + \boldsymbol{\delta_{reply}}$
- Then $$\mathbf{ToF_{A_B}} = \frac{\boldsymbol{\delta_{loop}} - \boldsymbol{\delta_{reply}}}{2}$$



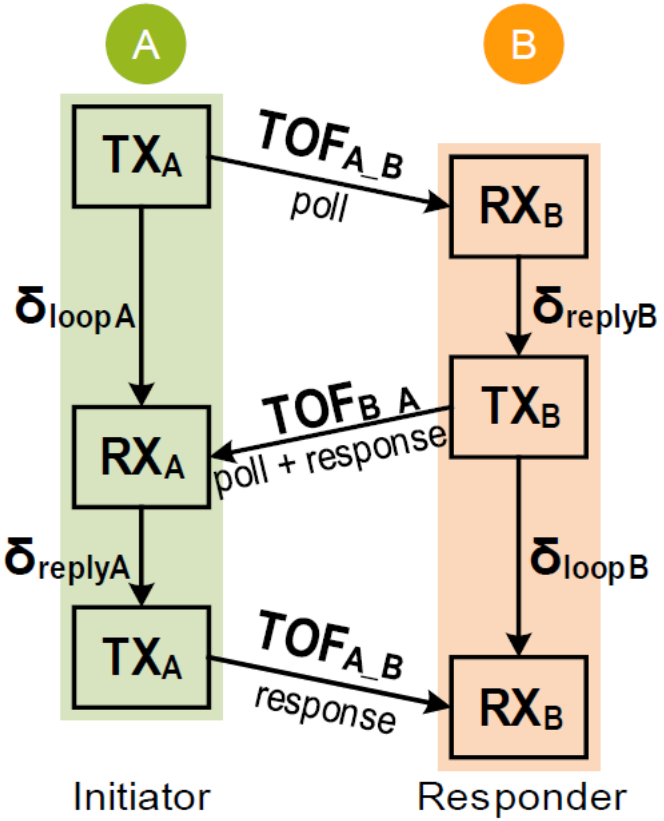
RANGING TECHNICS

Double Sided – Two Way Ranging (DS-TWR)

- SS-TWR is a simple technic, but any clock reference difference between Initiator and responder leads to TOF calculation error
- To avoid this **Two Way Ranging Double Sided** (DS-TWR) could be used
- Two SS-TWR are performed
- Each device can calculate the ToF

- Then $$ToF_{A_B} = \frac{\delta_{loopA} \times \delta_{loopB} - \delta_{replyA} \times \delta_{replyB}}{\delta_{loopA} + \delta_{loopB} + \delta_{replyA} + \delta_{replyB}}$$

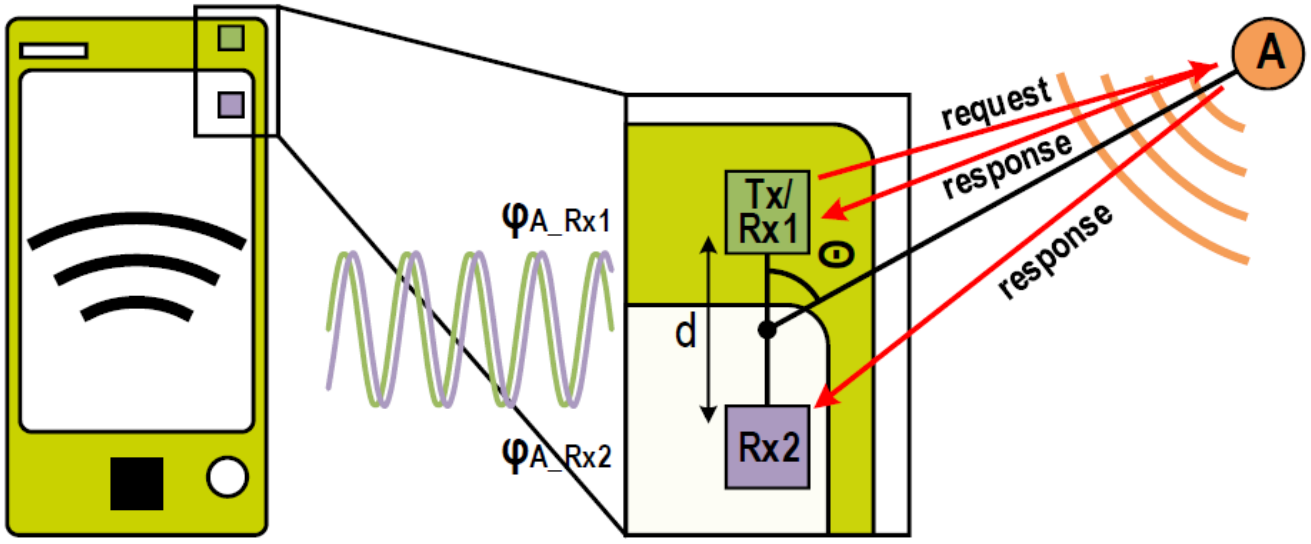
- Clock difference in the two devices are compensated but more energy is required due to 3 frames
- **-High accuaracy**



RANGING TECHNICS

Angle Of Arrival (AOA) – Phase Difference Of Arrival

- **Angle of Arrival (AoA)** is based on **Phase Difference of Arrival (PDoA)** method.
- **PDOA** calculation is done during response of TWR using the phase difference (**Δφ**) between two antennas of received signals.
- Phase difference (**Δφ**) is calculated using carrier signal and not UWB pulse envelope.
- Reception antenna spacing (**d**) should be below **λ/2**. (~18,8mm for CH9)
- **AoA** (**θ**) is a computation of above quantities:



$$\Delta\varphi = \varphi_{A_Rx1} - \varphi_{A_Rx2}$$
$$= 2\pi \times f \times \Delta t$$
$$= 2\pi \times f \times \frac{\Delta D}{c}$$
$$= 2\pi \times \frac{d \times \cos(\theta)}{\lambda}$$

$$\Rightarrow \theta = \text{acos}\left(\frac{\Delta\varphi \times \lambda}{2\pi \times d}\right)$$

03
UWB PHY

UWB PHY

Packet Format

There two type of packet in UWB Ranging:

- SP3: contain STS (encrypted/secured)
- SP0: contain user defined data

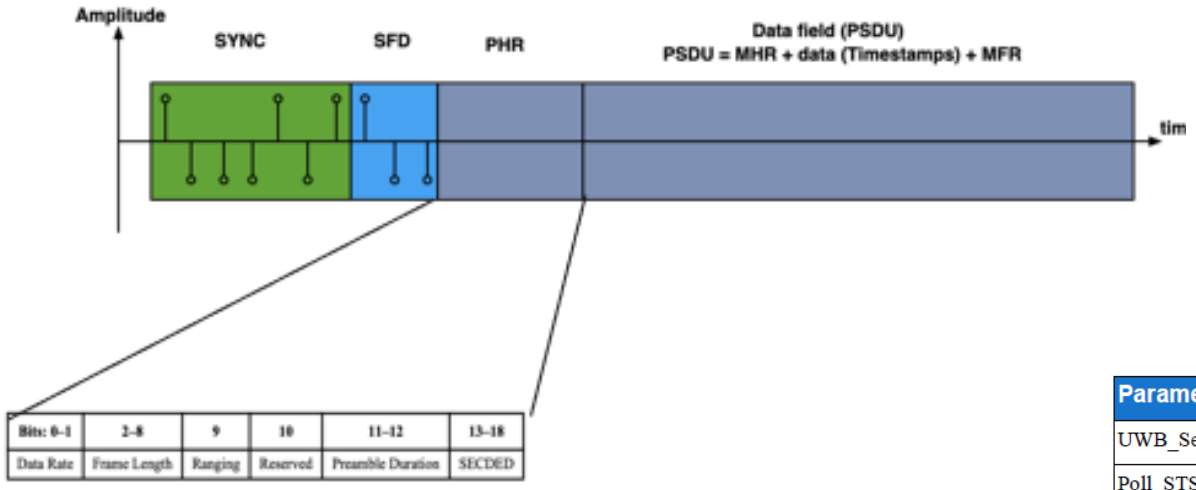


Figure 21-2: SP3

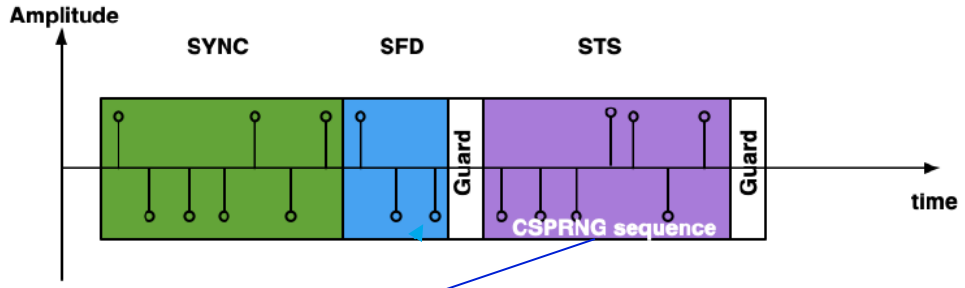


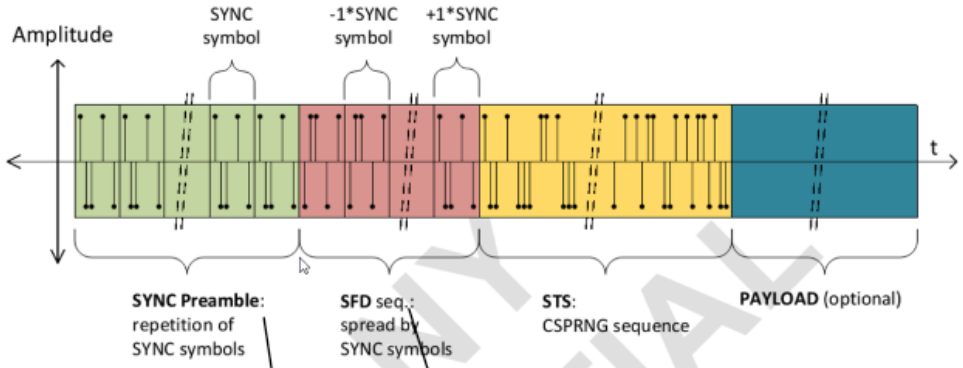
Table 20-4: The definition of parameters in the Pre-Poll Message.

Parameters	Length (bytes)	Value	Description
UWB_Session_ID	4	0 - (2^{32}-1)	ID of the UWB ranging session
Poll_STS_Index	4	0 - (2^{31}-1)	STS index of the succeeding POLL message

Details in the file CCC-TS-101-Digital-Key-R3-1.2.2-APPROVED_ChapterPHY Chapter 21

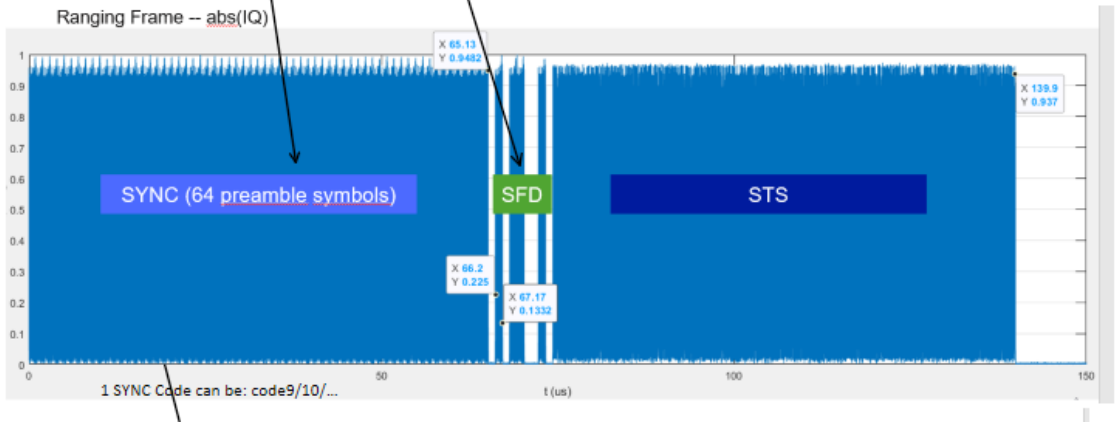
UWB PHY

Packet Format



SFD start of frame delimiter:
Defined in the context of the IEEE802.15.4-2011 [1] standard. Indicates the completion of the preamble section of the frame

STS Secure Training Sequence:
The transfer of payload for authentication is not required, because the intended recipient knows upfront which STS sequence to expect and can use the STS forauthentication.



Details in the file CCC-TS-101-Digital-Key-R3-1.2.2-APPROVED_ChapterPHY Chapter 21

UWB PHY

Packet Format



21.3.2.1 Precursor-Free Pulse – PulseShape 0x2

PulseShape 0x2 is a special case within PulseShape 0x1, by applying minimum-phase conversion of PulseShape 0x0 using method described in [34]. A Python implementation of the linear-to-minimum phase conversion is included in Appendix G, and FIR samples for a span of 8 chips at oversampling of 32x and β of 0.45 for PulseShape 0x2 are again included in Appendix H, both provided as informative references. It is recommended to use PulseShape 0x2 whenever possible if transmitter chooses to implement precursor-free pulse shape.

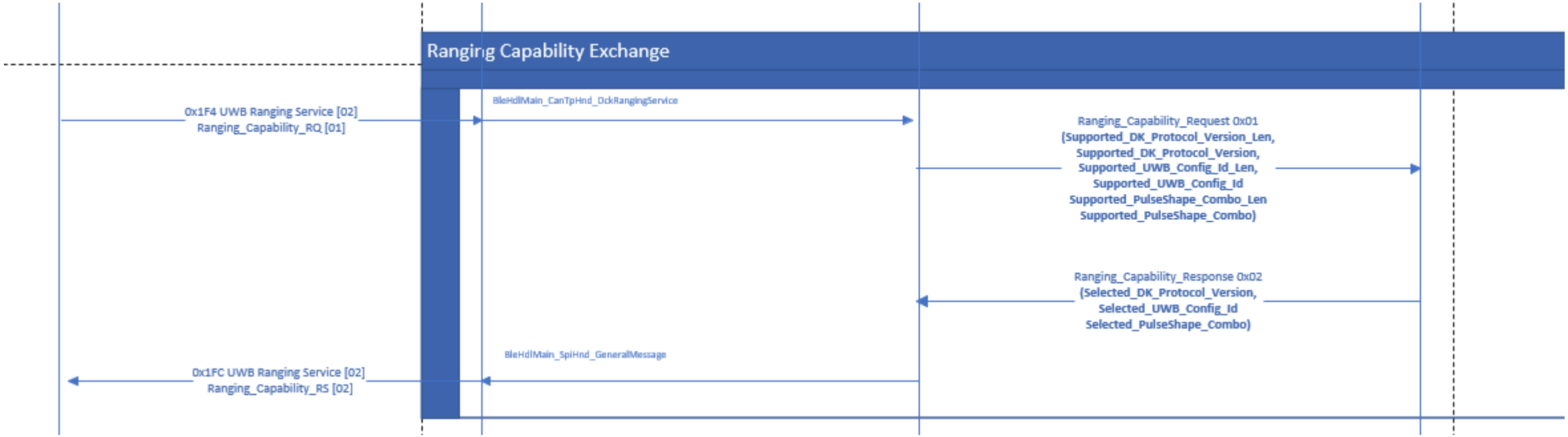
Pulse Shape	Description	bitfield
Symmetrical Root-Raised-Cosine Pulse	symmetrical Root-Raised-Cosine Pulse, referred to as PulseShape 0x0, the RF envelope (or equivalent baseband pulse amplitude) is expressed by [31]	0x0
Precursor-Free Pulse	group of pulse shapes that meet the time-domain mask of Fig. 89 in [33].	0x1
	a special case within PulseShape 0x1, by applying minimum-phase conversion of PulseShape 0x0 using method described in [34].	0x2

Table 21-3: Summary of pulse shapes

Details in the file CCC-TS-101-Digital-Key-R3-1.2.2-APPROVED_ChapterPHY Chapter 21

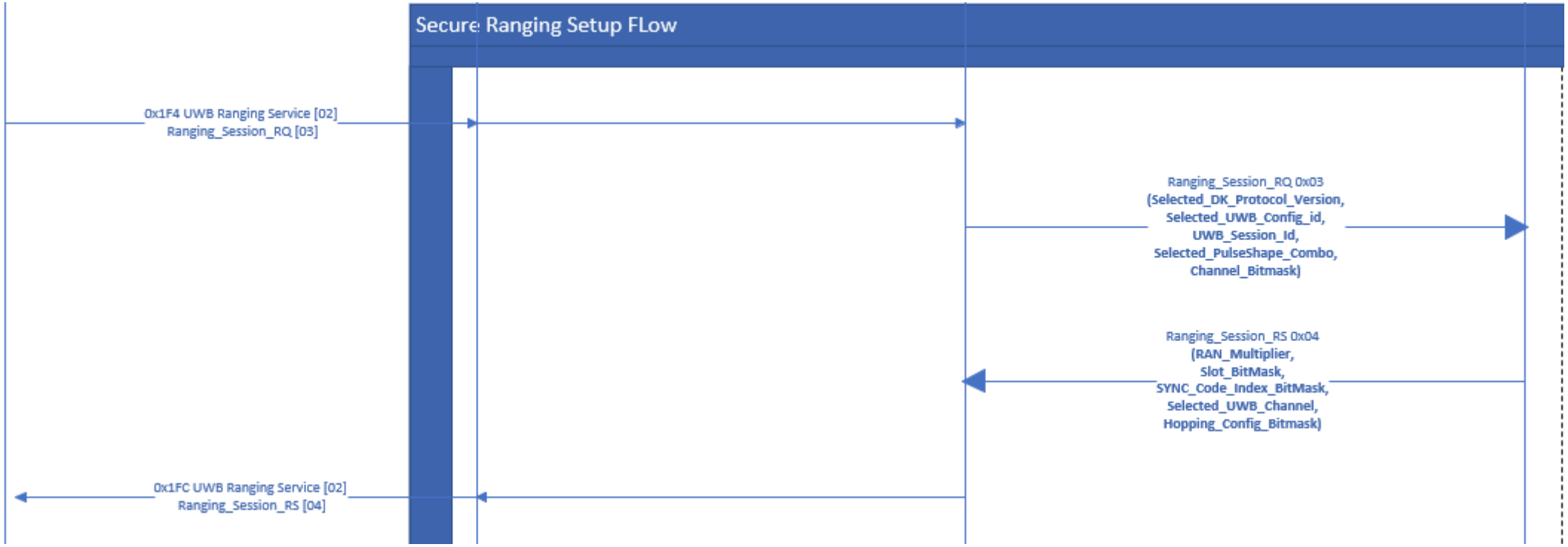
04
RANGING SESSION SETUP

RANGING SESSION SETUP



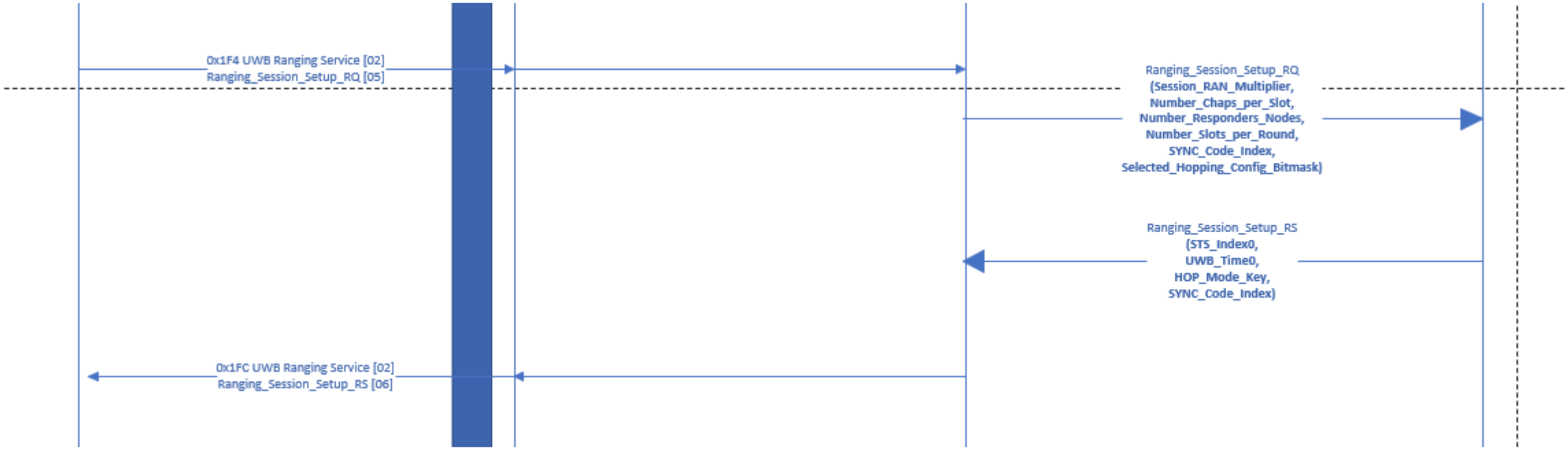
Details in the file CCC-TS-101-Digital-Key-R3-1.2.2-APPROVED_ChapterPHY Chapter 20

RANGING SESSION SETUP



Details in the file CCC-TS-101-Digital-Key-R3-1.2.2-APPROVED_ChapterPHY Chapter 20

RANGING SESSION SETUP



Details in the file CCC-TS-101-Digital-Key-R3-1.2.2-APPROVED_ChapterPHY Chapter 20

05
MAC TIME GRID

MAC TIME GRID

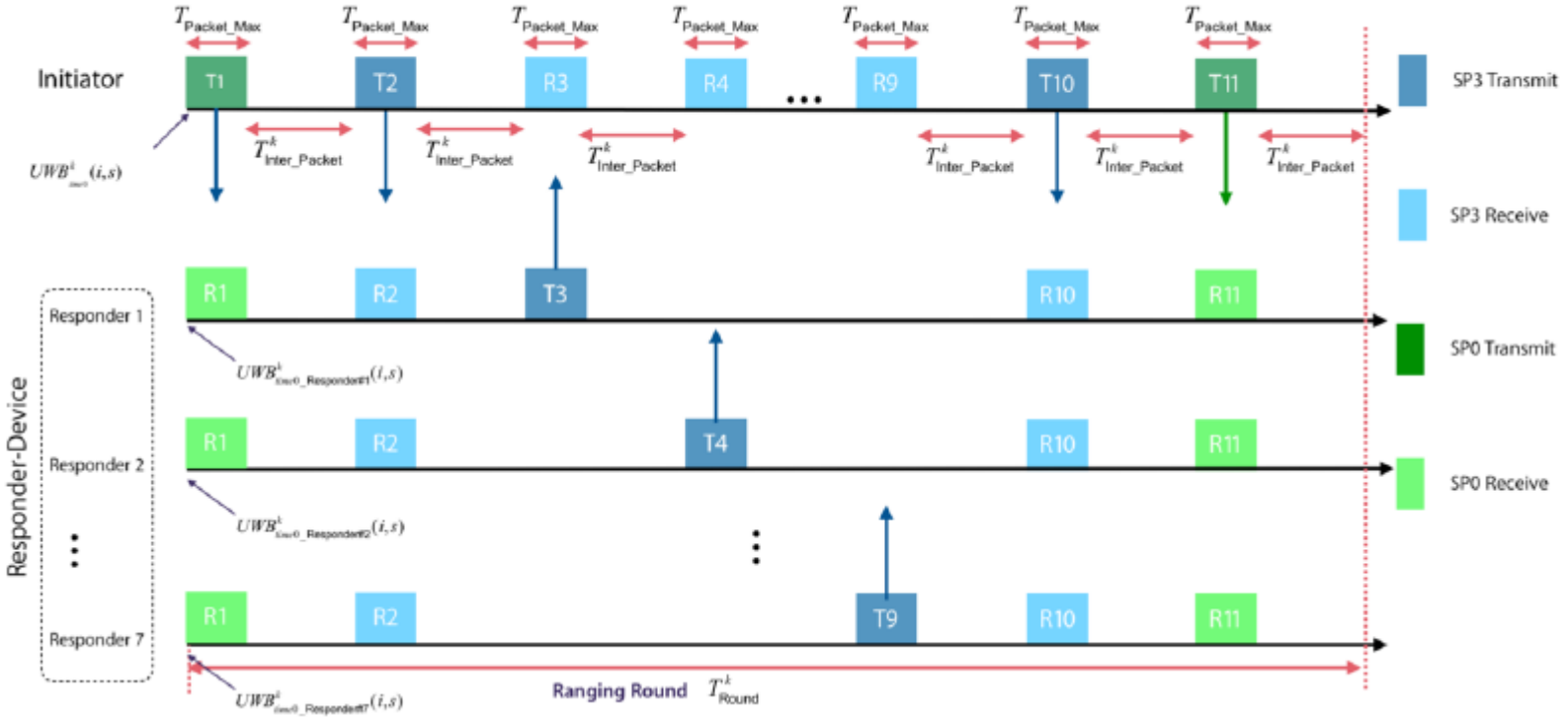
Ranging Exchange Sequence

The DK UWB ranging protocol is a one-to-many (O2M) ranging protocol between the initiator and the responder-device

The protocol uses a ranging block structure where each ranging block is T^k_{Block} in time duration

Each ranging block is divided into N^k_{Round} ranging rounds that are long enough to carry out one complete ranging exchange

Figure 20-2: Digital Key One-to-Many Ranging Protocol.



Details in the file CCC-TS-101-Digital-Key-R3-1.2.2-APPROVED_ChapterPHY Chapter 20

MAC TIME GRID

Ranging Exchange Sequence

Each Ranging Message should be executed in its time slot

T1:

- Initiator send msg PREPOLL to all Responders
- Responder open Rx window to scan incoming msg

T2:

- Initiator send msg POLL to all Responders
- Responder open Rx window to scan incoming msg

T3:

- Initiator open Rx window to scan incoming msg
- Responder1 send msg RESPONSE to Initiator

T4:

- Initiator open Rx window to scan incoming msg
- Responder2 send msg RESPONSE to Initiator

…

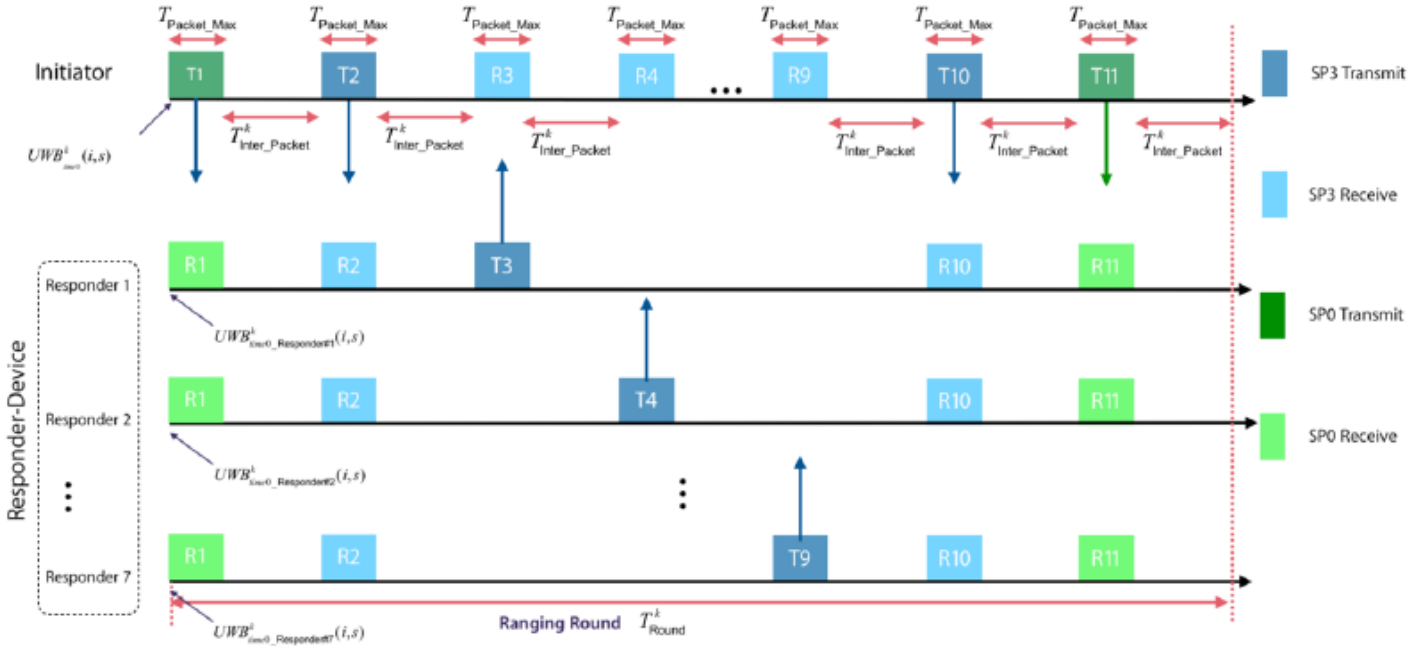
T10:

- Initiator send msg FINAL to all Responders
- Responder open Rx window to scan incoming msg

T11:

- Initiator send msg FINAL_DATA to all Responders
- Responder open Rx window to scan incoming msg

Figure 20-2: Digital Key One-to-Many Ranging Protocol.



Details in the file CCC-TS-101-Digital-Key-R3-1.2.2-APPROVED_ChapterPHY Chapter 20

MAC TIME GRID

Package Element

Table 20-3: Pre-Poll Request Message and its parameters.

UWB MAC Message	UWB MAC Message ID	Parameters
Pre-POLL	1	UWB_Session_ID, Poll_STS_Index, Ranging_Block, Hop_Flag, Round_Index

Table 20-4: The definition of parameters in the Pre-Poll Message.

Parameters	Length (bytes)	Value	Description
UWB_Session_ID	4	0 - (2^{32}-1)	ID of the UWB ranging session
Poll_STS_Index	4	0 - (2^{31}-1)	STS index of the succeeding POLL message
Ranging_Block	2	0 - 65535	Session's Index of current ranging block
Hop_Flag	1	0:No Hopping 1:Hopping	Hop flag for current ranging block as set from the ranging exchange in the previous ranging block For no hopping configuration this field is always 0 For continuous hopping configuration this field is always 1 (except for the first ranging block with index 0 after start or recovery of a ranging session).
Round_Index	2	0-65535	The ranging round index for the current ranging block as set from the ranging exchange in the previous ranging block. This field is set to 0 in first ranging block where we always start in ranging round 0.

Details in the file CCC-TS-101-Digital-Key-R3-1.2.2-APPROVED_ChapterPHY Chapter 20

MAC TIME GRID

Package Element

Table 20-5: Final_Data Message and its parameters.

UWB MAC Message	UWB MAC Message ID	Parameters
Final_Data	2	UWB_Session_ID, Ranging_Block, Hop_Flag, Round_Index, Final_STS_Index,

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UWB MAC Message	UWB MAC Message ID	Parameters
		Ranging_Timestamp_FINAL_TX, Number_Ranging_Responders, Responder_Index, Ranging_Timestamp_Responder_*l*, Ranging_Timestamp_Uncertainty_Responder_*l*, Ranging_Status_Responder_*l*, …

Table 20-6: The definition of parameters in the Final_Data Message.

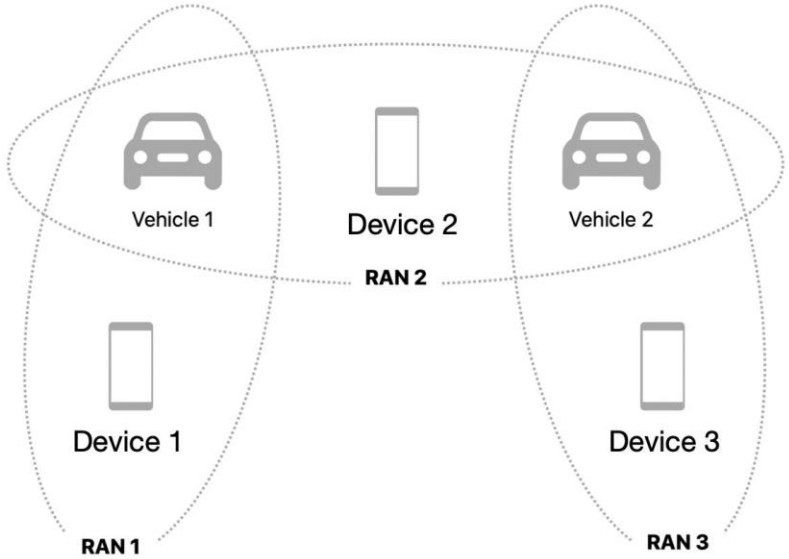
Parameters	Length (bytes)	Value	Description
UWB_Session_ID	4	0–(2^{32}-1)	ID of the UWB ranging session
Ranging_Block	2	0–65535	Index of current ranging Block
Hop_Flag	1	0: No Hopping 1: Hopping	Hop Flag for next ranging Block For no hopping configuration this field is always 0 For continuous hopping configuration this field is always 1
Round_Index	2	0 – 65535	The ranging round in which the ranging cycle will be executed in the next ranging block
Final_STS_Index	4	0 – (2^{31}-1)	STS index of the preceding Final message.
Ranging_Timestamp_FINAL_TX	4	Range = 0 to 0XFFFFFFFF Timestamp = N × 1/128 × 1/499.2e6 sec Time Range = 0 to 67.21 ms	Time difference between POLL and Final messages transmit times at the initiator.
Number_Ranging_Responders	1	0–255	Number of timestamps to follow in this message
Responder_Index	1	0–255	Index of Responder *l* whose timestamp data is referred to in the Ranging_Timestamp_Responder_*l*, Ranging_Timestamp_Uncertainty_Responder_-*l*, and Ranging_Status_Responder_*l* fields
Ranging_Timestamp_Responder_*l*	4	Range = 0 to 0XFFFFFFFF Timestamp=N × 1/128 × 1/499.2e6 sec Time Range = 0 to 67.21 ms	Time difference between POLL and RESPONSE of Responder *l*, as received by the initiator.
Ranging_Timestamp_Uncertainty_-Responder_*l*	1	See Section 6.9.1.7 in [33]	Range of values from 1.5 cm–3.6 m at various confidences
Ranging_Status_Responder_*l*	1	See Table 20-7	The status for the response frame from the responder. See the corresponding description for the ranging status in Table 20-7
…	…	…	Repeat parameters Responder_Index to Ranging_Status_Responder_*l* for each responder whose time stamp data has been added (up to Number_Ranging_Responders times)

MAC TIME GRID

Ranging session: An initiator and a responder-device engaged in a continuous ranging procedure

Ranging area network (RAN): An initiator and its (one or more) ranging sessions

- A single RAN shall have one initiator only and may have more than one responder - device (e.g. a device ranging with two vehicles).
- Each responder-device may have a different number of logical responders
- A single responder-device may belong to multiple RANs (e.g. a single vehicle ranging with two different devices)

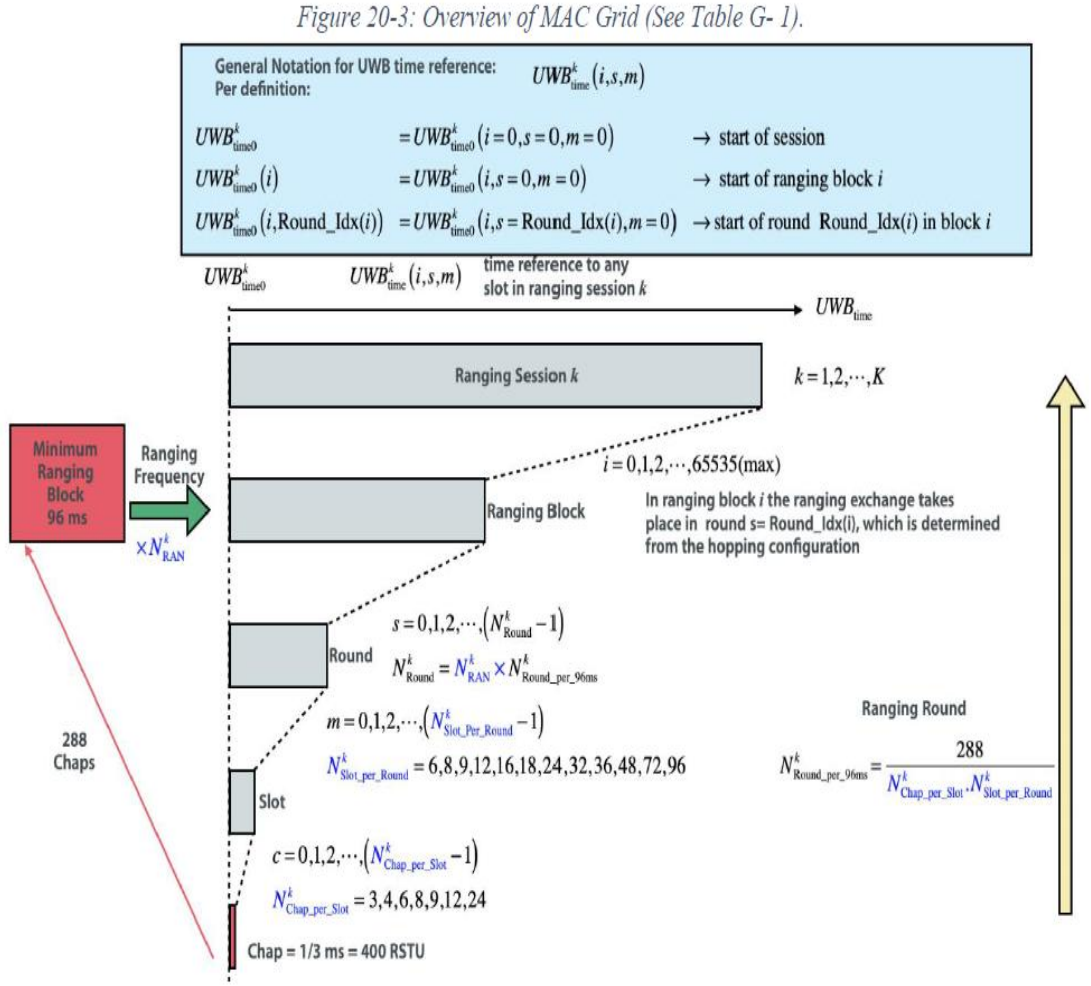


MAC TIME GRID

One RAN consists of one initiator and k = 1, …, K responder-devices

The k-th ranging session (which is associated with the k-th responder-device) is identified with a unique UWB_ Session_Id^k

The k-th ranging session is defined relative to a specific time reference . This time reference is defined by the initiator with respect to its clock base UWB^k_{time0}. This time reference defines the beginning of a series of consecutive ranging blocks



Details in the file CCC-TS-101-Digital-Key-R3-1.2.2-APPROVED_ChapterPHY Chapter 20

MAC TIME GRID

Session:

- Ranging Session of k-th Responder- Device

Block:

- Defined time for Ranging exchange between Initiator**(s)** and Responder – Device
- $T_{block} = N_{RAN} * 96ms$ (96ms is minimum time)

Round:

- Defined time for Ranging exchange between Initiator and all Responder – Device
- Ranging exchange can be executed in any Round in one Block
- $T_{round} = T_{slot} * N_{slot_per_round}$; $N_{slot_per_round} = 6,8,9,12,16,18,24,32,36,48,72,96$

Slot:

- Defined time when the message should be sent or when the receiver should open Rx window
- $T_{slot} = T_{chap} * N_{chap_per_slot}$; $N_{chap_per_slot} = 3,4,6,8,9,12,24$

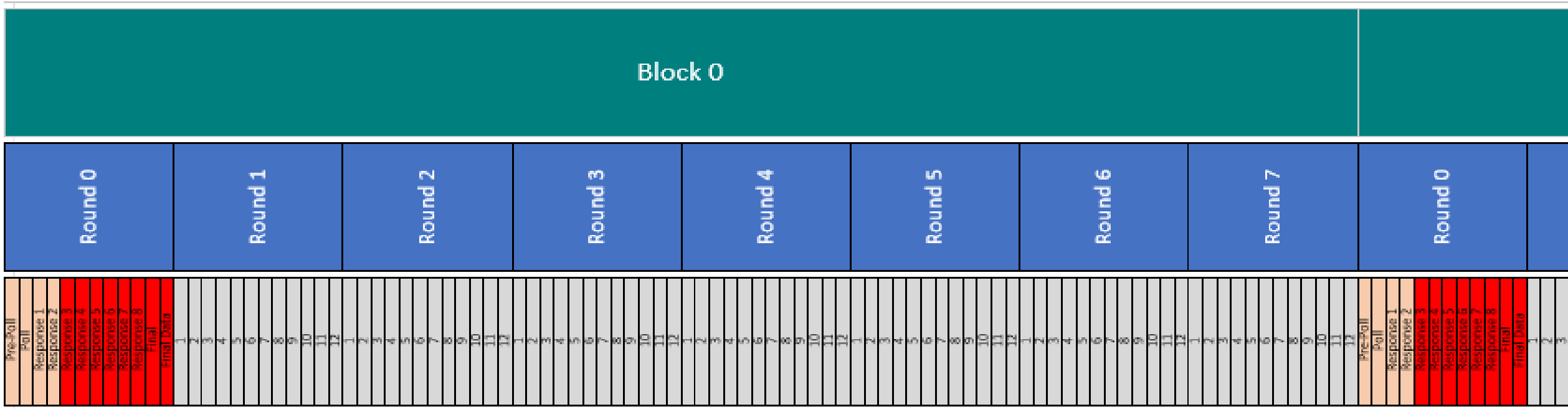
Chap:

- Minimum time unit in Ranging.
- Chap = 1/3 ms

Detail information (CCC document) can be found in the following slide

Details in the file CCC-TS-101-Digital-Key-R3-1.2.2-APPROVED_ChapterPHY Chapter 20

MAC TIME GRID



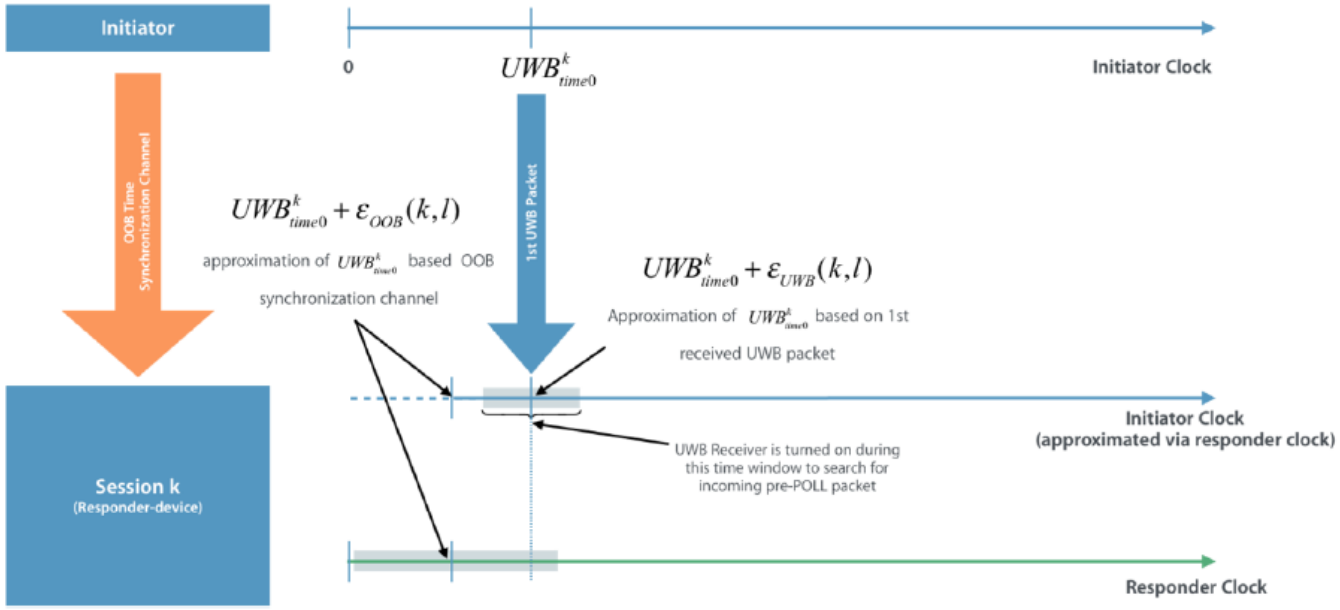
06
MAC TIME GRID SYNCHRONIZATION

MAC TIME GRID SYNCHRONIZATION

Each ranging session is defined relative to a specific time reference UWB^{k}_{time0} which is defined relative to the initiator's internal clock. The time reference UWB^{k}_{time0} establishes the MAC time grid for the k-th ranging session.

The MAC time grid shall be estimated, with reasonable accuracy, at each responder (at the responder-device). This allows each responder to determine when to expect to receive the Pre-Poll message from the initiator, when to send its response message back to the initiator, and when to expect to receive the Final and Final_Data messages from the initiator

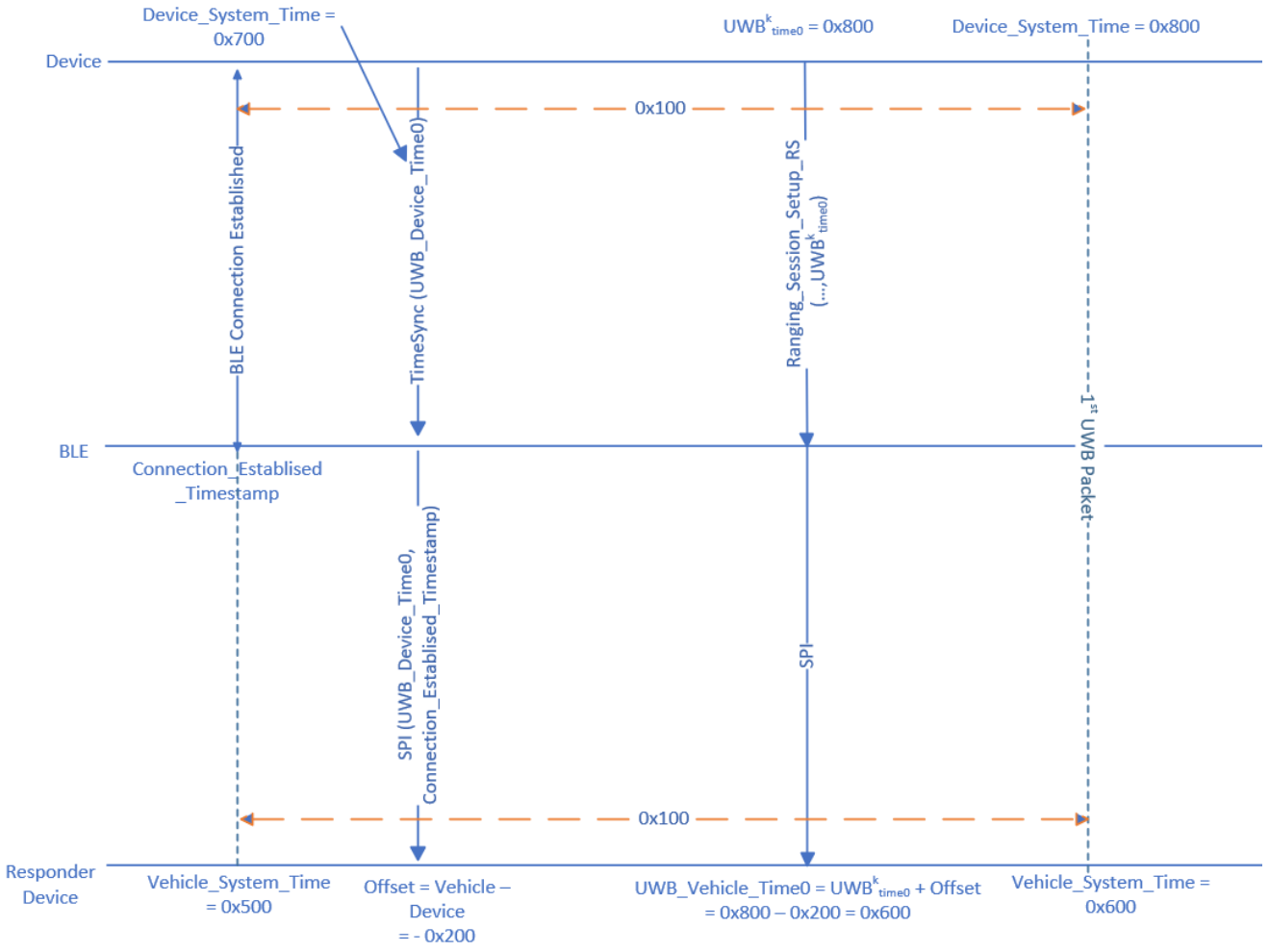
Figure 20-4: Overview of Block Synchronization Approach.



Details in the file CCC-TS-101-Digital-Key-R3-1.2.2-APPROVED_ChapterPHY Chapter 20

MAC TIME GRID SYNCHRONIZATION

1. Initiator send internal clock (Device_System_Time) to Responder via TimeSync Msg
2. Responder calculate clock deviation between itself and Initiator
3. Initiator send UWB_{time0} to Responder via RangingSessionSetup_RS to announce at which timestamp of Initiator's clock, Ranging should be started
4. Responder calculate starting time based on clock deviation and UWB_{time0}



07
HOPPING AND RESOURCE MANAGEMENT

HOPING AND RESOURCE MANAGEMENT

Intra-RAN Resource Conflict:

- A resource conflict may occur when the initiator would have to serve two different ranging exchanges (to two different responder-devices) at the same time.
- This scenario may be resolved at the initiator by prioritizing one of the ranging sessions involved. The target ranging round may be used for the ranging session with the highest priority and the initiator shall skip the round for the other ranging sessions

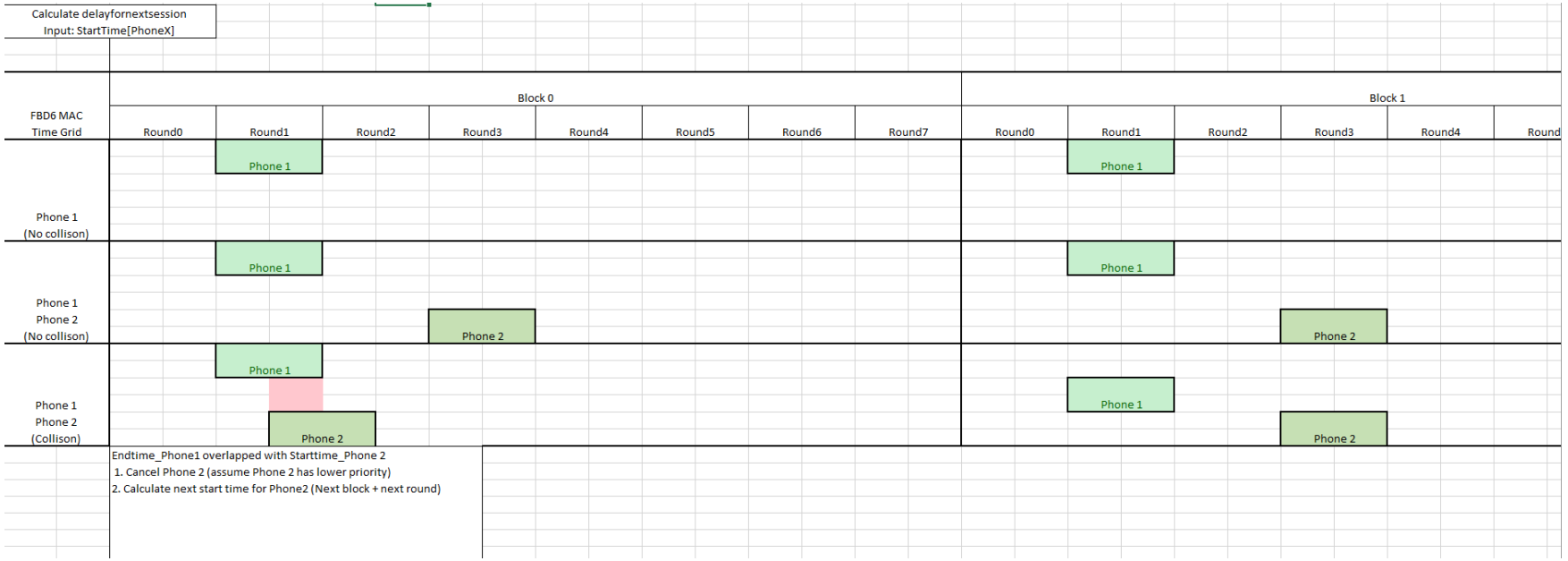
Inter-RAN Resource Conflict:

- A resource conflict occurs when a responder would have to serve ranging rounds from two (or more) different RANs at the same time
- This scenario may be resolved by prioritizing one RAN and skipping the round for the other RANs. The prioritization is determined by the involved responder-device

HOPING AND RESOURCE MANAGEMENT

Hopping mode: Depend on Hopping flag, Ranging exchange can be occurred at specific Round

- No hopping: Round_Idx(i+1) = Round_Idx(i) = 0 (no resource conflict)
- Continous hopping: Round_Idx(i+1) = S_{i+1}(no resource conflict)
- Adaptive hopping:
  - No resource conflict: No hopping
  - Resource conflict: Continous hopping

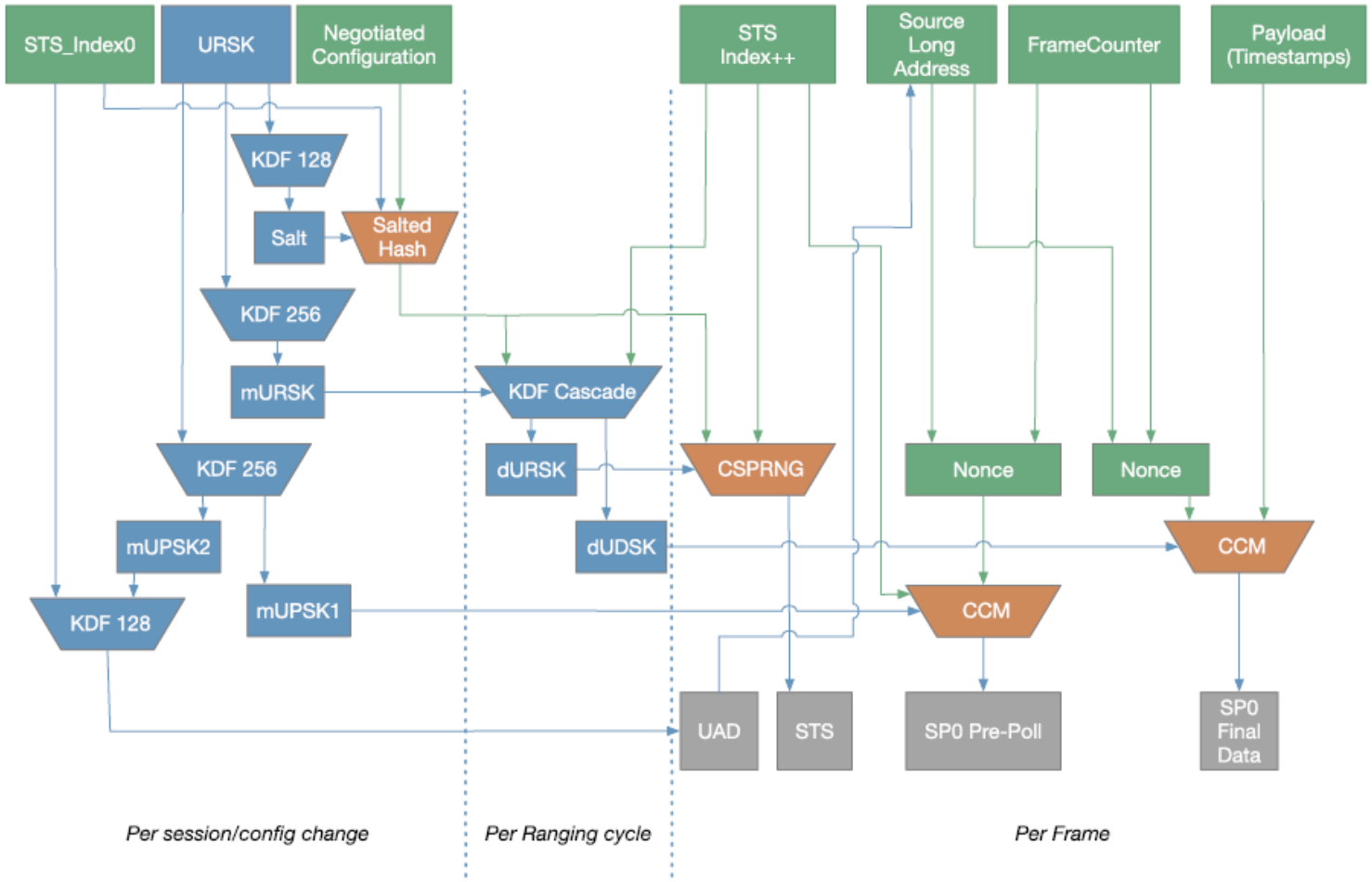


08
UWB CRYPTO SEQUENCE

UWB CRYPTO SEQUENCE

Overview

Figure 22-1: Overview of the KDF used and its relationship.



Details in the file CCC-TS-101-Digital-Key-R3-1.2.2-APPROVED_ChapterPHY Chapter 22

UWB CRYPTO SEQUENCE

SP0 encryption

STS_index encryption (SP0 Pre-POLL)

- *Algorithm*: AES-CCM* as described in Section 9.3 of [31]
- *Mode*: Encrypted and authenticated [31](Security Level 6 - ENC-MIC64)

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- *K (Key)* = mUPSK1[0:128]
  - *N (Nonce)* = Source Long Address | Frame Counter | Nonce Security Level as described in Section 9.3.2.1 of [31]
  - Incrementing Frame Counter, STS_Index in payload

Note: The key mUPSK1 is static for the ranging session.

Timestamps encryption (SP0 Final_Data timestamp frame)

- Algorithm: AES-CCM* per as described in Section 9.3 of [31]
- Mode: Encrypted and authenticated [31](Security Level 6 - ENC-MIC64)
- K (Key) = dUDSK, where dUDSK is defined later in this subsection
- N (Nonce) = Source Address | Frame Counter | Nonce Security Level as described in Section 9.3.2.1 of [31]
- Incrementing Frame Counter

Note: The probability of IV/Key repetition is insignificant since the key dUDSK shall be updated for every ranging block.

Details in the file CCC-TS-101-Digital-Key-R3-1.2.2-APPROVED_ChapterPHY Chapter 22

UWB CRYPTO SEQUENCE

Example of SaltedHash's KDF

22.1.3.2 SaltedHash Key Derivation

Purpose: Preserve integrity of ranging configuration

Occurrence: Once per ranging configuration negotiation

Cryptography:

Salt[0:128] = CMAC(URSK, 0x00000001 || **0x53414C54** || 0x00 || 0x000000 || 0x00000080)

CMAC= AES256-CMAC as PRF, r = 32, h=128

Reference: See Section 5.1 of [4] KDF in Counter Mode

PaddedSalt[0:256] = 0x000..0000 || Salt

RangingConfiguration = Selected_Protocol_Version || Selected_UWB_Config_Id || UWB_Session_Id || STS_Index0 || Number_Responder_Nodes || Session_RAN_Multiplier || Number_Slot_per_Round || Number_Chaps_per_Slot || Selected_PulseShape_Combo

SaltedHash = CMAC(PaddedSalt, RangingConfiguration)

CMAC = AES256-CMAC as PRF, r = 32, h=128

Note: ASCII (**0x53414C54**) = "SALT"

RangingConfiguration parameters are fully described in Section 20.5.2.

Details in the file CCC-TS-101-Digital-Key-R3-1.2.2-APPROVED_ChapterPHY Chapter 22